

FUNDACIÓN FULGOR

CMOS Course

Homework N°4

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1 Introduction

This report analyzes and calculates the *Differential Gain* of the circuit shown in Fig. 1.

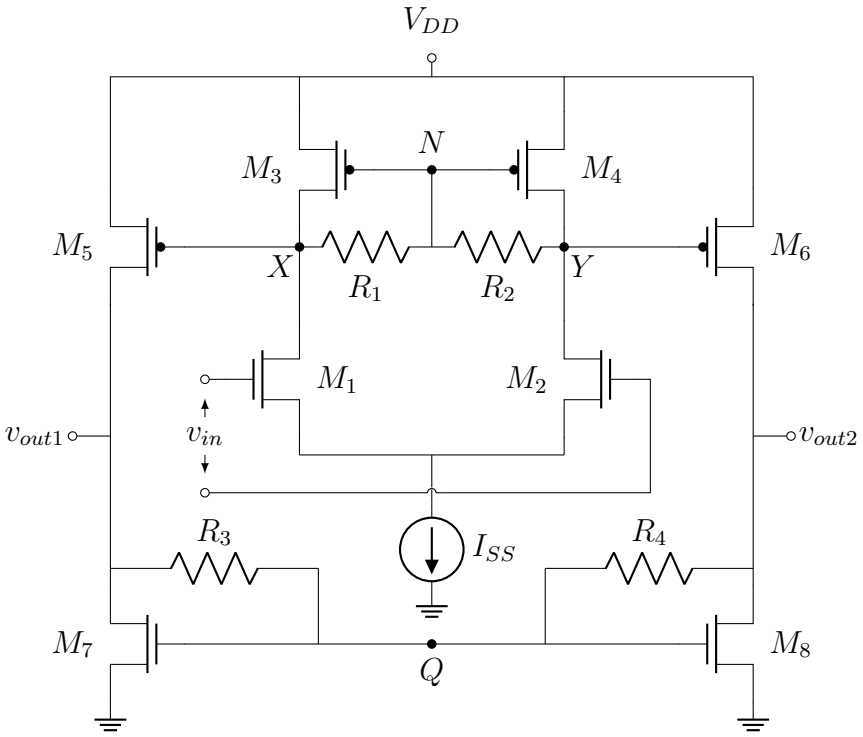


Figure 1: Circuit proposed to analyze.

The report will present details of the step-by-step calculation of the differential voltage gain under the assumptions that: Channel-length modulation is negligible ($\lambda = 0$) for all transistors and thus $r_0 = \infty\,[\Omega]$; R_1 is equal to R_2, and R_3 is equal to R_4 and all transistors have the same W/L relationship.

2 Circuit Analysis

The circuit shown in Fig. 1 consist of a two-stage fully differential *CMOS* amplifier:

- The first stage comprises a differential pair (M_1 and M_2) biased by a tail current source I_{SS}. The load for this initial stage consists of a symmetric network formed by resistors R_1 and R_2 alongside active *PMOS* devices M_3 and M_4.
- The second stage is composed of common-source amplifiers (M_5 and M_6) loaded by an output network consisting of resistors R_3 and R_4, and *NMOS* current sources M_7 and M_8.

The following subsections will analyze these two stages separately based on the half circuit model proposed by Bezhad Razavi. [1] Note that since the circuit is perfectly sym-

metric and we are evaluating purely differential signals, the nodes connecting symmetric components become AC virtual grounds.

2.1 First Stage Gain

This first stage is a differential pair composed of M_1 and M_2 biased with the tail current source I_{SS} as mentioned in section 2. Note that the input signal is applied differentially, meaning:

$$\begin{cases} v_{in1} = \frac{v_{in}}{2} \\ v_{in2} = -\frac{v_{in}}{2} \end{cases} \tag{1}$$

The midpoint between R_1 and R_2 (labeled as node N in Fig. 1) is an AC virtual ground. Consequently, the gates of M_3 and M_4 are tied to zero AC voltage. Since the sources of M_3 and M_4 are connected to V_{DD}, which is also an AC ground, we have that:

$$v_{GS3} = v_{GS4} = 0\,[V] \tag{2}$$

With $\lambda = 0$, the output resistance of these transistors is infinite ($r_{o3} = r_{o4} = \infty\,[\Omega]$). Therefore, the equivalent impedance looking into the load at node X is simply the resistor R_1 connected to the AC virtual ground such that the small-signal voltage at this node is given by:

$$v_X = -g_{m1} \cdot R_1 \cdot \frac{v_{in}}{2} \tag{3}$$

This result can be verified with the *Example 4.12* of Bezhad Razavi's book: *Design of Analog CMOS Integrated Circuits*, as shown in Fig. 2.

Example 4.12

It is possible to obviate the need for V_b in the circuit of Fig. 4.37(b) as shown in Fig. 4.38(a), where R_1 and $R_2(= R_1)$ are relatively large. In the absence of signals, $V_X = V_Y = V_N = V_{DD} - |V_{GS3,4}|$. That is, M_3 and M_4 are "self-biased." Determine the differential voltage gain of this topology.

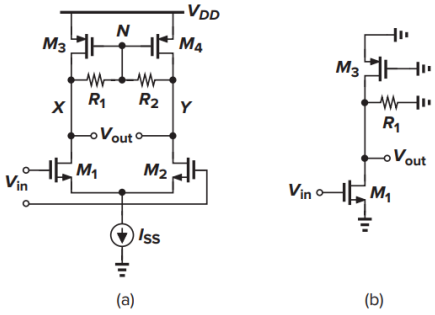


Figure 4.38

Solution

For differential outputs, V_N does not change (why?) and can be considered ac ground. Shown in Fig. 4.38(b), the half-circuit yields

$$|A_v| = g_{m1}(r_{O1} \| R_1 \| r_{O3}) \tag{4.63}$$

If the resistors are much greater than $r_{O1}||r_{O3}$, then they negligibly reduce the gain. ◀

Figure 2: Stage 1 Gain verification. [2, p. 124]

2.2 Second Stage Gain

The second stage consists of M_5 operating as a *Common-Source PMOS* amplifier driven by its input voltage v_X. The load network at the output node V_{out1} consists of R_3 and M_7.

Similar to the first stage, the midpoint between R_3 and R_4 (node Q) acts as an AC virtual ground for differential signals. This places the gate of M_7 at AC ground, resulting in $v_{GS7} = 0$.

Once again, given that $\lambda = 0$, M_7 acts as an ideal open circuit for small signals ($r_{o7} = \infty\,[\Omega]$) and thus, the equivalent load for M_5 is therefore just the resistor R_3. Therefore, the output voltage at v_{out1} is given by Eq. 4.

$$v_{out1} = -g_{m5} \cdot R_3 \cdot v_X \tag{4}$$

Replacing v_X in Eq. 4 with the result obtained in Eq. 3 we have:

$$v_{out1} = -g_{m5} \cdot R_3 \cdot \left(-g_{m1} \cdot R_1 \cdot \frac{v_{in}}{2}\right) \tag{5}$$

$$v_{out1} = g_{m1} \cdot g_{m5} \cdot R_1 \cdot R_3 \cdot \frac{v_{in}}{2} \tag{6}$$

By symmetry, the voltage at the complementary output V_{out2} is:

$$v_{out2} = -g_{m2} \cdot g_{m6} \cdot R_2 \cdot R_4 \cdot \frac{v_{in}}{2} \tag{7}$$

where given the assumptions made in section 1, another way of writing Eq. 7 is:

$$v_{out2} = -g_{m1} \cdot g_{m5} \cdot R_1 \cdot R_3 \cdot \frac{v_{in}}{2} \tag{8}$$

2.3 Total Differential Gain

The total differential output voltage is given by:

$$v_{out,diff} = v_{out1} - v_{out2} \tag{9}$$

where substituting Eq. 6 and Eq. 8 into Eq. 9, we have:

$$v_{out,diff} = g_{m1} \cdot g_{m5} \cdot R_1 \cdot R_3 \cdot v_{in} \tag{10}$$

Finally, we have that the differential voltage gain is:

$$A_v = \frac{v_{out,diff}}{v_{in}} = g_{m1} \cdot g_{m5} \cdot R_1 \cdot R_3 \tag{11}$$

3 Conclusion

The derived differential voltage gain obtained in Eq. 11, demonstrates a highly coherent physical behavior for this two-stage topology. Under purely differential excitation, the nodes driving the Common-Mode Feedback ($CMFB$) transistors (M_3, M_4, M_7, and M_8) act as AC virtual grounds, effectively disabling these active devices in the small-signal regime. Coupled with the $\lambda = 0$ constraint, which yields infinite intrinsic output resistances, all parasitic loading effects vanish entirely. This logically reduces each amplification stage to an ideal transconductor driving solely its respective $CMFB$ sensing resistor. Thus, the overall performance perfectly reflects a cascade of two unattenuated common-source amplifiers.

4 Bibliography

[1] B. Razavi, *Fundamentals of Microelectronics*, 2nd Edition. John Wiley & Sons, 2013, ISBN: 978-1-118-15632-2.

[2] B. Razavi, *Design of Analog CMOS Integrated Circuits*, 2nd Edition. Mc Graw Hill Education, 2016, ISBN: 978-0-07-252493-2.